Contents

1 Joint, Conditional and Relative Entropy **3**

1.1 Joint Entropy . . . 3

1.2 Mutual Information . . . 5

1.3 Conditional Entropy . . . 6

1.3.1 Simple Intuition . . . 6

1.3.2 Relation with Joint Entropy . . . 6

1.4 Relative Entropy . . . 7

1.4.1 Mutual Information as Relative Entropy . . . 8

Chapter 1. Joint, Conditional and Relative Entropy

1.1 Joint Entropy

Joint Entropy

The joint entropy $H(X,Y)$ of a pair of random variables (X,Y) with joint distribution $P(x,y)$ is the average uncertainty associated with the pair of outcomes taken together.

Joint Entropy Formula

$$H(X,Y) = -\sum_{x\in X}\sum_{y\in Y} P(x,y)\log_2 P(x,y)$$
$$= -E\log_2 P(x,y)$$

- $H(X,Y)$ = joint entropy of the variables X and Y
- $P(x,y)$ = joint probability of the pair (x,y)
- $E[\cdot]$ = expectation

Example 1.1

Two variables are considered for a student:

- X = study hours
- Y = attendance

Their joint probabilities are listed below. Show that the variables are dependent on each other.

X	**Y**	**P(X, Y)**
Low	Low	0.20
Low	High	0.10
High	Low	0.15
High	High	0.55

Step 1: Joint entropy contribution of each case

$$H_{LL}(X,Y) = -0.2\log_2(0.2) = 0.46$$
$$H_{LH}(X,Y) = -0.1\log_2(0.1) = 0.33$$
$$H_{HL}(X,Y) = -0.15\log_2(0.15) = 0.41$$
$$H_{HH}(X,Y) = -0.55\log_2(0.55) = 0.4743$$

$$H(X,Y) = 0.46 + 0.33 + 0.41 + 0.4743 = 1.6743$$

Step 2: Marginal entropy for study hours

$$P_L = 0.2 + 0.1 = 0.3, \qquad P_H = 0.15 + 0.55 = 0.7$$

$$H(X) = -\left[0.3\log_2(0.3) + 0.7\log_2(0.7)\right] = 0.881$$

Step 3: Marginal entropy for attendance

$$P_L = 0.2 + 0.15 = 0.35, \qquad P_H = 0.1 + 0.55 = 0.65$$

$$H(Y) = -\left[0.35\log_2(0.35) + 0.65\log_2(0.65)\right] = 0.934$$

Conclusion

$$H(X) + H(Y) = 0.881 + 0.934 = 1.815$$

Since $H(X) + H(Y) > H(X, Y)$, the variables X and Y are dependent. We can also say that the variables contain mutual information.

1.2 Mutual Information

Mutual Information

Mutual information measures the amount of information that one random variable X provides about another random variable Y. It quantifies the reduction in uncertainty about one variable when the other is known.

Mutual Information Formula

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

$$\text{or equivalently,} \qquad I(X;Y) = H(X) - H(X \mid Y)$$

- $I(X;Y)$ = mutual information between X and Y
- $H(X)$, $H(Y)$ = marginal entropies
- $H(X,Y)$ = joint entropy
- $H(X \mid Y)$ = conditional entropy of X given Y

Example 1.2

Calculate the joint entropy for the following data and also find the mutual information.

Y \ X	**0**	**1**
0	$\frac{1}{2}$	$\frac{1}{4}$
1	$\frac{1}{8}$	$\frac{1}{8}$

Step 1: Joint entropy

$$\begin{aligned} H(X,Y) &= -\left[\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{4}\log_2\frac{1}{4} + \frac{1}{8}\log_2\frac{1}{8} + \frac{1}{8}\log_2\frac{1}{8}\right] \\ &= 1.75 \end{aligned}$$

Step 2: Entropy of Y

$$P(Y=0) = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}, \qquad P(Y=1) = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$$

$$H(Y) = \frac{3}{4}\log_2\frac{4}{3} + \frac{1}{4}\log_2 4 = 0.811$$

Step 3: Entropy of X

$$P(X=0) = \frac{1}{2} + \frac{1}{8} = \frac{5}{8}, \qquad P(X=1) = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}$$

$$H(X) = \frac{5}{8}\log_2\frac{8}{5} + \frac{3}{8}\log_2\frac{8}{3} = 0.954$$

Step 4: Conditional entropies

$$H(X \mid Y) = H(X,Y) - H(Y) = 1.75 - 0.811 = 0.939$$

$$H(Y \mid X) = H(X,Y) - H(X) = 1.75 - 0.954 = 0.796$$

Step 5: Mutual information

$$I(X;Y) = H(X) + H(Y) - H(X,Y) = 0.954 + 0.811 - 1.75 = 0.015$$

1.3 Conditional Entropy

Conditional Entropy

Conditional entropy measures the amount of uncertainty remaining in a random variable when another related variable is already known. It tells us how much additional information is still needed.

1.3.1 Simple Intuition

Consider the set $\{1, 2, 3, 4, 5, 6\}$. If we only know whether the outcome is odd or even, some uncertainty still remains.

- If the outcome is **odd**, the possible values are $\{1, 3, 5\}$.
- If the outcome is **even**, the possible values are $\{2, 4, 6\}$.

This remaining uncertainty is the idea captured by conditional entropy.

Conditional Entropy Formula

$$\begin{aligned} H(Y \mid X) &= -\sum_{x \in X} P(x) \sum_{y \in Y} P(y \mid x) \log_2 P(y \mid x) \\ &= -\sum_{x \in X} \sum_{y \in Y} P(x, y) \log_2 P(y \mid x) \end{aligned}$$

- $H(Y \mid X)$ = uncertainty of Y after X is known
- $P(y \mid x)$ = conditional probability of $Y = y$ given $X = x$
- $P(x, y)$ = joint probability of X and Y

1.3.2 Relation with Joint Entropy

Starting from the joint entropy,

$$\begin{aligned} H(X, Y) &= -\sum_{x \in X} \sum_{y \in Y} P(x, y) \log_2 P(x, y) \\ &= -\sum_{x \in X} \sum_{y \in Y} P(x, y) \log_2 \big(P(x) P(y \mid x)\big) \\ &= -\sum_{x \in X} \sum_{y \in Y} P(x, y) \log_2 P(x) - \sum_{x \in X} \sum_{y \in Y} P(x, y) \log_2 P(y \mid x) \\ &= -\sum_{x \in X} P(x) \log_2 P(x) - \sum_{x \in X} \sum_{y \in Y} P(x, y) \log_2 P(y \mid x) \end{aligned}$$

Hence,

$$H(X, Y) = H(X) + H(Y \mid X)$$

Similarly,

$$H(X, Y) = H(Y) + H(X \mid Y)$$

1.4 Relative Entropy

Relative Entropy

Relative entropy, also called Kullback-Leibler divergence, measures how different an estimated probability distribution is from the true distribution.

Kullback-Leibler Divergence

$$D(P \parallel Q) = \sum_{x \in X} P(x) \log_2 \frac{P(x)}{Q(x)}$$

- $P(x)$ = true probability mass function
- $Q(x)$ = estimated probability mass function
- $D(P \parallel Q)$ = relative entropy from Q to P

Basic Points

- If $P(x) = Q(x)$, then $D(P \parallel Q) = 0$.
- In the note's illustration:

$$D(P \parallel P) = 0, \qquad D(P \parallel A) = 0.25, \qquad D(P \parallel B) = 1.25$$

- Higher relative entropy means the estimated distribution deviates more from the actual distribution.

Example 1.3

Consider a scenario where the true distribution is

$$P_{\text{dog}} = 1, \qquad P_{\text{cat}} = 0$$

For model A,

$$P'_{\text{dog}} = 0.1, \qquad P'_{\text{cat}} = 0.9$$

For model B,

$$P'_{\text{dog}} = 0.9, \qquad P'_{\text{cat}} = 0.1$$

Find the relative entropy using KL divergence.

For model A:

$$D(P \parallel P'_A) = P_{\text{dog}} \log_2 \frac{P_{\text{dog}}}{P'_{\text{dog}}} + P_{\text{cat}} \log_2 \frac{P_{\text{cat}}}{P'_{\text{cat}}}$$

$$= 1 \log_2 \frac{1}{0.1} + 0 \cdot \log_2 \frac{0}{0.9} = \log_2 10 = 3.32 \text{ unit}$$

For model B:

$$D(P \parallel P'_B) = P_{\text{dog}} \log_2 \frac{P_{\text{dog}}}{P'_{\text{dog}}} + P_{\text{cat}} \log_2 \frac{P_{\text{cat}}}{P'_{\text{cat}}}$$

$$= 1 \log_2 \frac{1}{0.9} + 0 \cdot \log_2 \frac{0}{0.1} = 0.15 \text{ unit}$$

So model B is better because it has lower KL divergence.

1.4.1 Mutual Information as Relative Entropy

For two dependent random variables (X, Y), mutual information can be written as the relative entropy between the joint distribution $P(x, y)$ and the product of the marginals $P(x)P(y)$:

Mutual Information from Relative Entropy

$$I(X;Y) = \sum_{x,y} P(x,y) \log_2 \frac{P(x,y)}{P(x)P(y)} = D\big(P(x,y) \,\|\, P(x)P(y)\big)$$

$$\begin{aligned}
I(X;Y) &= \sum_{x,y} P(x,y) \log_2 \frac{P(x)P(y \mid x)}{P(x)P(y)} \\
&= \sum_{x,y} P(x,y) \log_2 \frac{P(y \mid x)}{P(y)} \\
&= \sum_{x,y} P(x,y) \log_2 P(y \mid x) - \sum_{x,y} P(x,y) \log_2 P(y) \\
&= -H(Y \mid X) + H(Y)
\end{aligned}$$

So,

$$I(X;Y) = H(Y) - H(Y \mid X)$$

Also,

$$H(X,Y) = H(X) + H(Y \mid X)$$

Therefore,

$$H(Y \mid X) = H(X,Y) - H(X)$$

Substituting this into the previous expression,

$$I(X;Y) = H(Y) - H(X,Y) + H(X)$$

Thus the mutual information formula becomes

$$I(X;Y) = H(X) + H(Y) - H(X,Y)$$

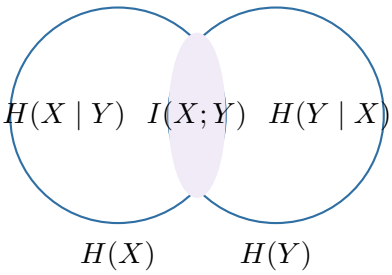

Figure 1.1: Mutual information as the overlap between $H(X)$ and $H(Y)$